ACADEMIC PROJECT REPORT & VIVA GUIDE

STYLE-RENT PLATFORM

Online Luxury Clothing, Formals & Costumes Rental Solution

PROJECT OVERVIEW & SPECIFICATIONS

- System Type: Full-Stack React 19 + TypeScript + Vite + Tailwind CSS SPA
- Core Domain: E-Commerce / Quick-Commerce Fashion Rental System
- Specialized Features: 1-Day Rental, 2-Hour SOS Express, StylePass 0% Deposit, Live GPS Tracking
- Target Metro Coverage: Mumbai, Delhi NCR, Bengaluru, Hyderabad, Chennai, Kolkata, Pune
- Live Demo URL: https://ais-pre-bhqatkonulyfu5bb5sbu4s-698741090226.asia-southeast1.run.app

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ACADEMIC SUBMISSION NOTE

Prepared for academic viva examination, technical demo presentation, and institutional project submission.

Includes complete source code reference, GitHub repository commands, and architectural proofs.

1. EXECUTIVE SUMMARY & PROBLEM STATEMENT

Occasion wear (royal sherwanis, designer bridal lehengas, 3-piece tuxedo suits, and mythological/theatrical costumes) suffers from an acute price-to-utility mismatch: a premium designer outfit costing Rs. 25,000 to Rs. 1,00,000 is worn only 1-2 times before remaining idle in wardrobes. Style-Rent transforms this underutilized asset class through a circular rental economy. Customers rent verified, dry-cleaned designer apparel starting at Rs. 599 to Rs. 899 with 0% deposit (StylePass), flexible durations (1-day, 3-day, 7-day, 10-day), and same-day or 2-hour SOS emergency doorstep fulfillment.

2. TECHNICAL ARCHITECTURE & COMPONENT TOPOLOGY

Frontend Framework	React 19 (SPA Architecture) with TypeScript (~5.7)
Build System & Dev Server	Vite 6 with Hot Module Replacement & Tree-Shaking
Styling & Design System	Tailwind CSS v4 with bespoke Indian luxury palette (Maroon, Royal Gold, Sand)
Iconography & Visual Assets	Lucide React (16+ contextual SVG icons)
State Management	React Hooks (useState, useMemo, useEffect) with persistent LocalStorage
Business Logic & Math	Dedicated modular utilities (rentalCalculations.ts) for duration pricing & deposits
Delivery Simulation Engine	Live animated multi-step rider progression with real-time ETA & Delivery OTP
Operations Dashboard	Role-based administrative portal: inventory management, dispatches, and return QA

3. DATA FLOW & LIFECYCLE DIAGRAM

Step 1: User Browsing & Filters
Filters by gender, occasion, rental duration, and instant same-day delivery.

Step 2: Date Selection & Fitting
Calendar algorithm auto-calculates return dates. Standard sizing or custom tailoring.

Step 3: Smart Cart & Deposit Logic
Calculates refundable security deposit or applies 0% deposit waiver with StylePass.

Step 4: Express Checkout & OTP
Selects Standard (Free), Same-Day (Rs. 99), or 2-Hour SOS (Rs. 299). Generates 4-digit OTP.

Step 5: Live GPS Delivery Tracker
Interactive tracker tracks rider progress, distance, vehicle speed, and contact info.

Step 6: Post-Rental QA & Dry Cleaning
Operations portal logs outfit return, checks for fabric stains, and clears deposit.

3. COMPLETE FOLDER-WISE DIRECTORY & FILE MAP

/public	Static assets, generated project ZIP archives, and sample image models.
/src/App.tsx	Root application state orchestrator: cart, wishlist, orders, active modals, and tabs.
/src/types/index.ts	Strongly typed contracts: Product, Order, CartItem, UserMembership, RentalDuration.
/src/utils/rentalCalculations.ts	Pricing formulas: 1-day (65%), 7-day (1.6x), 10-day (2.1x), SOS fee (Rs. 299), deposit waiver.
/src/data/mockData.ts	Initial catalog data, reviews, StylePass tiers, metro cities list, and default orders.
/src/components/Navbar.tsx	Sticky top navigation with category dropdowns, search bar, StylePass, and Track button.
/src/components/HeroSection.tsx	High-conversion hero banner displaying trust pillars, SOS badges, and value proposition.
/src/components/ProductCard.tsx	Catalog cards showing dynamic pricing per selected duration, discount, and quick rent.
/src/components/ProductDetailModal.tsx	Full product modal: size selection, custom tailoring measurements, and fabric details.
/src/components/RentalCalendar.tsx	Date range picker with booking constraints, weekend highlighting, and blackout dates.
/src/components/CartDrawer.tsx	Sliding cart sheet calculating rental total, delivery surcharge, and deposit summary.
/src/components/CheckoutModal.tsx	Doorstep checkout form: speed selector (Standard, Same-Day, 2-Hr SOS), payment method.
/src/components/OrderConfirmationModal.tsx	Order success screen with rental timeline, delivery OTP, and link to Live GPS Tracker.
/src/components/DeliveryTrackerModal.tsx	Live GPS simulation with moving courier, speed indicators, rider details, and OTP verification.
/src/components/MembershipModal.tsx	StylePass Royalty Club tiers (Silver, Gold, Black) offering 0% security deposits.
/src/components/AdminDashboard.tsx	Internal portal for staff: Order fulfillment, live dispatch tracking, and return QA checks.
/src/components/ShareModal.tsx	Direct sharing modal with native share, WhatsApp, QR code, and ZIP download buttons.
/src/components/WishlistDrawer.tsx	Slide-over wishlist drawer allowing renters to bookmark preferred outfits.

4. MATHEMATICAL MODELS & BUSINESS ALGORITHMS

A. Multi-Duration Dynamic Pricing Matrix

- 1-Day Flash Event Rental: 65% of base 3-day rental fee (e.g., Rs. 899 base becomes Rs. 584)
- 3-Day Weekend Standard: 100% of base rental fee (Ideal for Sangeet, Wedding, or Corporate Summit)
- 7-Day Destination Wedding: 160% of base rental fee (Accommodates outstation travel days)
- 10-Day Festival / Theatrical Run: 210% of base rental fee (Costume production & mythological shoots)

B. Delivery Logistics Speed Tiers

- **Standard Doorstep (Free / Rs. 0):** Dispatched via premium courier 24-48 hours prior to reservation date.
- **Same-Day Guaranteed (+Rs. 99):** Dispatched via express hyperlocal fleet for same-day evening parties.
- **2-Hour SOS Emergency (+Rs. 299):** Priority rider assignment with live GPS tracking for last-minute wardrobe malfunctions.

C. StylePass Membership & 0% Deposit Security Architecture

- **Free Tier:** Standard refundable security deposit charged (Rs. 2,000 - Rs. 5,000) and refunded post-return QA.
- **Silver Member (Rs. 499/mo):** 50% deposit waiver + 1 free custom alteration session.
- **Gold Member (Rs. 999/mo):** 100% deposit waiver (Rs. 0 security deposit) + free same-day delivery upgrades.
- **Black Elite (Rs. 1,999/mo):** 100% deposit waiver + unlimited 2-hour SOS deliveries + dedicated wardrobe concierge.

D. Hygiene & Sanitization Protocol (Industry Benchmark)

1. 100% Hospital-Grade Eco-Dry Cleaning before and after every rental cycle.
2. High-pressure steam sanitization at 130 deg C to eliminate 99.9% of bacteria and allergens.
3. Double-layer breathable garment seal bag with tamper-evident reversible zipper for return courier.

5. GROUP OF 4 PRESENTATION DIVISION (5-7 MINUTES TOTAL)

Member 1: Intro & Problem Statement
Hook the audience with wardrobe cost dilemma. Present Style-Rent mission, value proposition, and catalog overview.

Member 2: User Journey & Custom Fitting
Walk through the live catalog: 1-day/3-day rental options, interactive calendar, and custom tailoring measurement inputs.

Member 3: Quick Commerce & Live GPS Tracker
Demonstrate checkout: 2-Hour SOS delivery (+Rs. 299), 4-digit delivery OTP, and live animated GPS rider tracking simulation.

Member 4: StylePass Membership & Admin Operations
Showcase StylePass 0% deposit club, financial sustainability, and the staff inspection/fulfillment admin portal.

6. TOP VIVA EXAMINATION QUESTIONS & HIGH-SCORING ANSWERS

Q1: How do you handle garment damages or unreturned items?

Ans: We implement a dual-shield strategy: Standard renters hold a pre-authorized security deposit. StylePass members are verified via KYC, and incidental wear-and-

Q2: Why choose React 19 and Tailwind v4 instead of traditional MPA?

Ans: Rental workflows require frictionless single-page interactivity (instant price recalculation on date change, modal sliders, live rider map polling) without disruptive fu

Q3: How does the system prevent date-booking overlaps for the same outfit?

Ans: The rental calendar computes inclusive ISO date intervals (startDate to returnDate). Admin inventory status tracks active dispatches, preventing double bookings

Q4: What is the unit economics and profitability model?

Ans: High-margin asset utilization: A designer sherwani purchased for Rs. 15,000 yields Rs. 1,800/rental. After 9 rentals, the asset breaks even and generates 70%+ g

7. GITHUB REPOSITORY SETUP & LOCAL COMMANDS

Step-by-Step GitHub Publishing Guide for User:

1. Create a new empty repository on your GitHub account named "style-rent" (https://github.com/new).
2. Open your project folder in terminal or VS Code and run the following commands:

   git init

   git add .

   git commit -m "Initial commit: Style-Rent luxury clothing rental platform"

   git branch -M main

   git remote add origin https://github.com/YOUR_GITHUB_USERNAME/style-rent.git

   git push -u origin main
3. Your code is now fully published to GitHub with complete commit history!

8. LOCAL DEVELOPMENT & DEMO RUNNER

Prerequisites: Node.js (v18 or higher) and npm installed.

Command 1: npm install **# Installs all required React 19, Lucide, and Vite modules**

Command 2: npm run dev **# Launches the development server on http://localhost:3000**

Command 3: npm run build **# Creates the optimized static production build in /dist**

PROJECT SUBMISSION CERTIFICATE & VERIFICATION

[OK] Project Title: Style-Rent: Online Luxury Clothing, Formals & Costumes Rental Platform

[OK] Key Technologies: React 19, TypeScript 5.7, Tailwind CSS v4, Lucide Icons, Vite 6

[OK] Production App Status: Compiled and Verified Live with Zero Warnings/Errors

[OK] Deliverables Included: Interactive Web App, Source Code ZIP, Full Project ZIP, PDF Viva Report

[OK] Direct Project Link: https://ais-pre-bhqatkonulyfu5bb5sbu4s-698741090226.asia-southeast1.run.app